

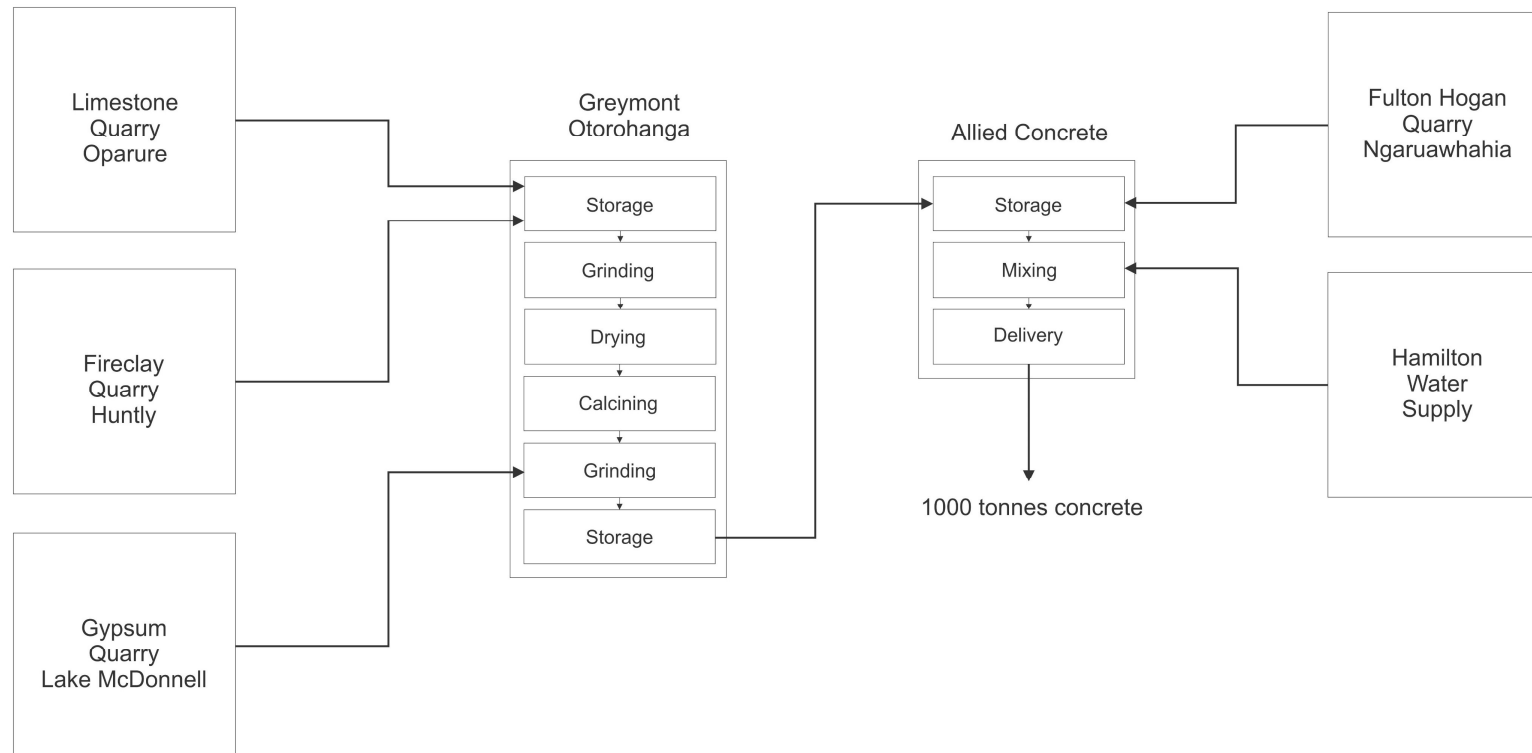
LCA 4.1 – Impact Analysis

ENGEN370

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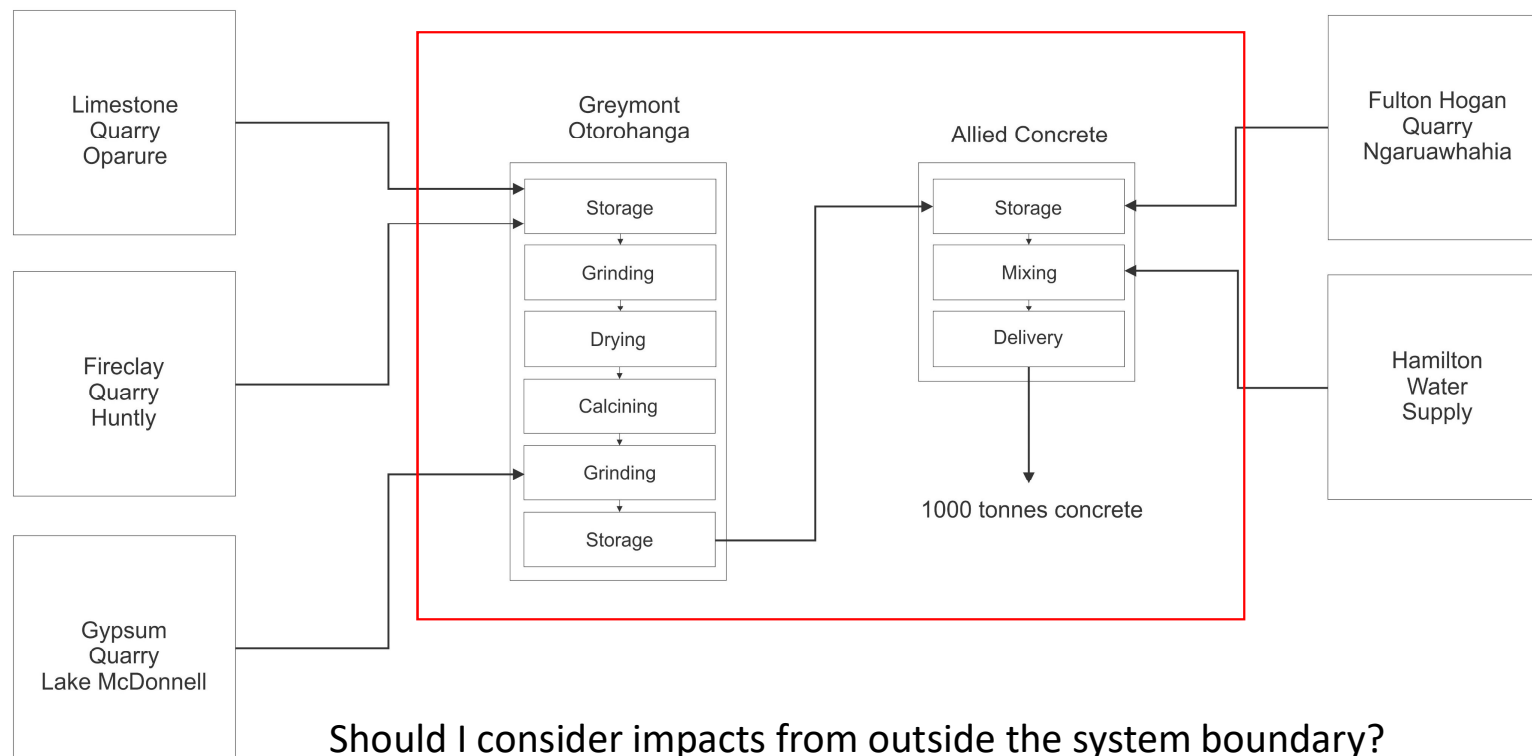
LCA - concrete manufacture

General flow diagram



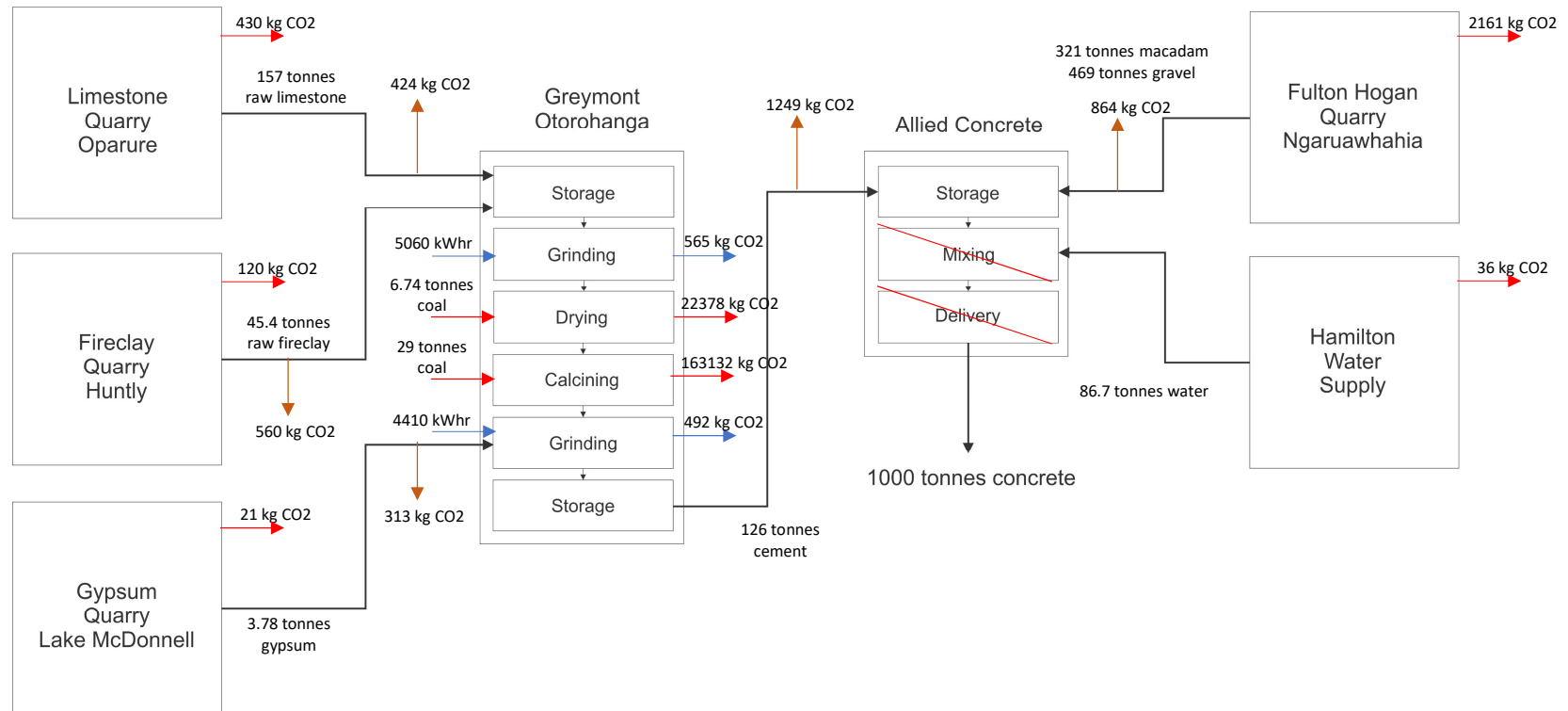
LCA - concrete manufacture

General flow diagram



LCA – concrete manufacture

“detailed” flow diagram



Categorising emission sources

- Use ISO 14064-1:2018 standard and the Greenhouse Gas Protocol for reporting and categorising emissions
- Emission sources organised into the following:
 - Scope 1 - Direct GHG emissions from sources owned or controlled by the company
 - Scope 2 - Indirect GHG emissions from the generation of purchased energy (in the form of electricity, heat or steam) that the organisation uses
 - Scope 3 - Other indirect GHG emissions occurring because of the activities of the organisation but generated from sources that it does not own or control (eg, air travel, shipping, trucking)

Scope	Category	Direct/Indirect emissions and removals	Source	New for this guide?
Scope 1	Category 1	Direct GHG emissions and removals	Fuel	
			Refrigerant and medical gases*	Yes
			Agriculture, forestry and other land uses	
Scope 2	Category 2	Indirect GHG emissions from imported energy	Purchased energy	
Scope 3	Category 3	Indirect GHG emissions from transportation	Business travel	
			Staff commute	Yes
			Freight transport	
			Refrigerant use (from chilled transport or air conditioner)	
	Category 4	Indirect GHG emissions from products an organisation uses	Transmission and distribution losses	
			Working from home	Yes
			Water supply and wastewater treatment	
			Materials and waste	
	Category 5	Indirect GHG emissions (use of products from the organisation)	Outside the scope of this guide	
	Category 6	Indirect GHG emissions (other sources)		

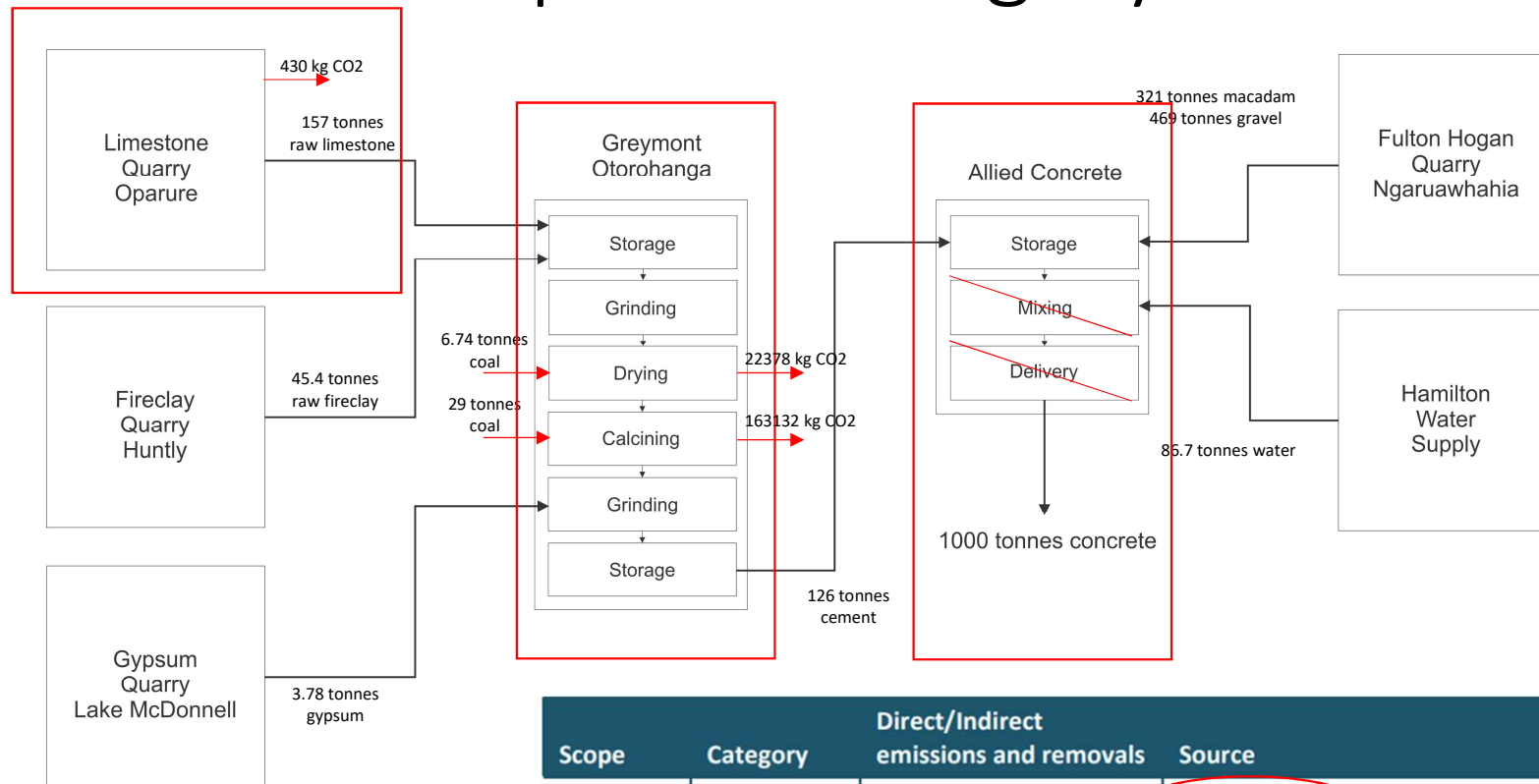
Table 2, Detailed Guide (MfE Guidance for Voluntary GHG Reporting – 2018), page 14

Categorising emission sources

- Lets assume:
 - Greymont and Allied Concrete are owned by the same company
 - The limestone quarry is owned by Greymont
 - The other quarries are owned by other companies
 - Shipping, rail and road transport is contracted out to other companies

Categorising emission sources

Scope 1 – Category 1

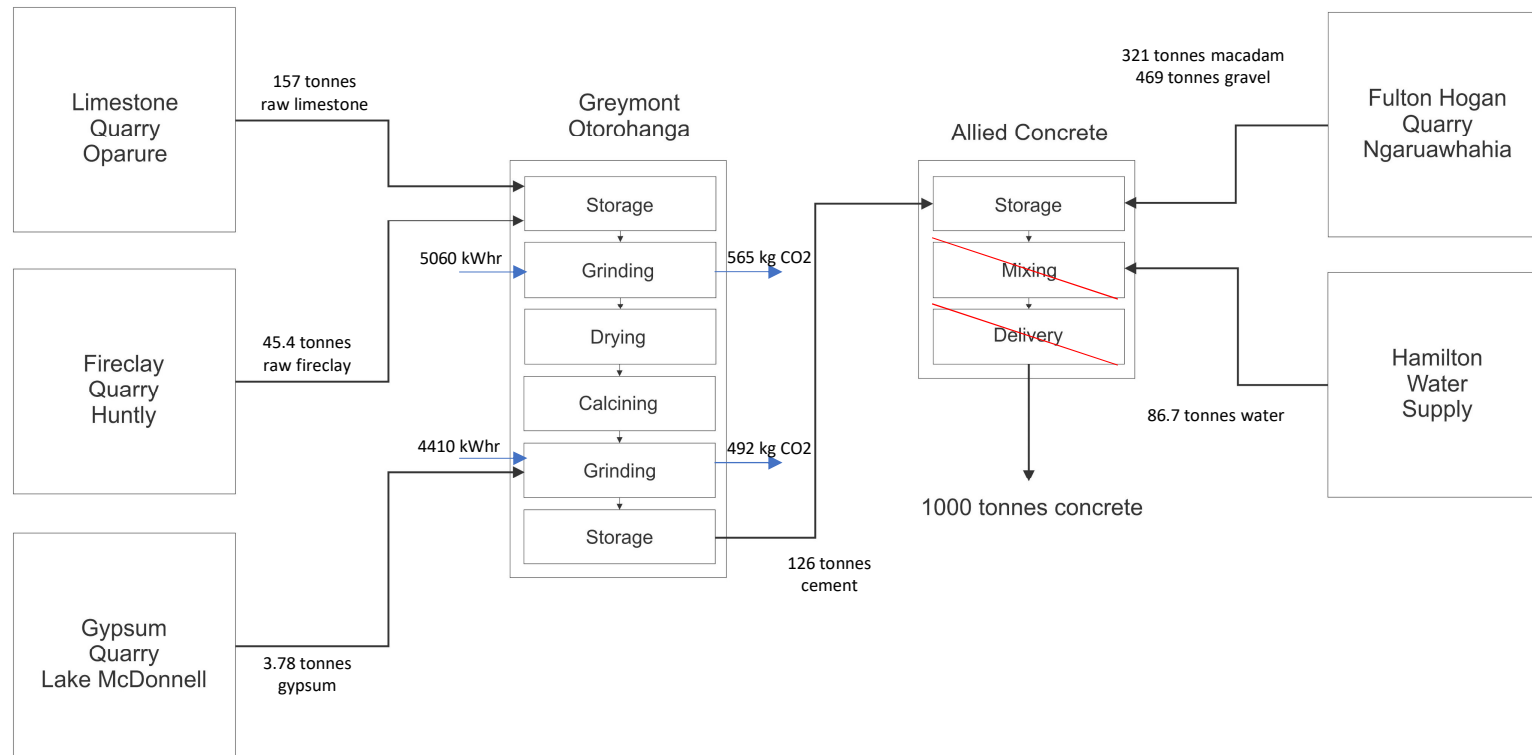


Scope 1 – sources owned by the company

Scope	Category	Direct/Indirect emissions and removals	Source
Scope 1	Category 1	Direct GHG emissions and removals	Fuel
			Refrigerant and medical gases*
			Agriculture, forestry and other land uses

Categorising emission sources

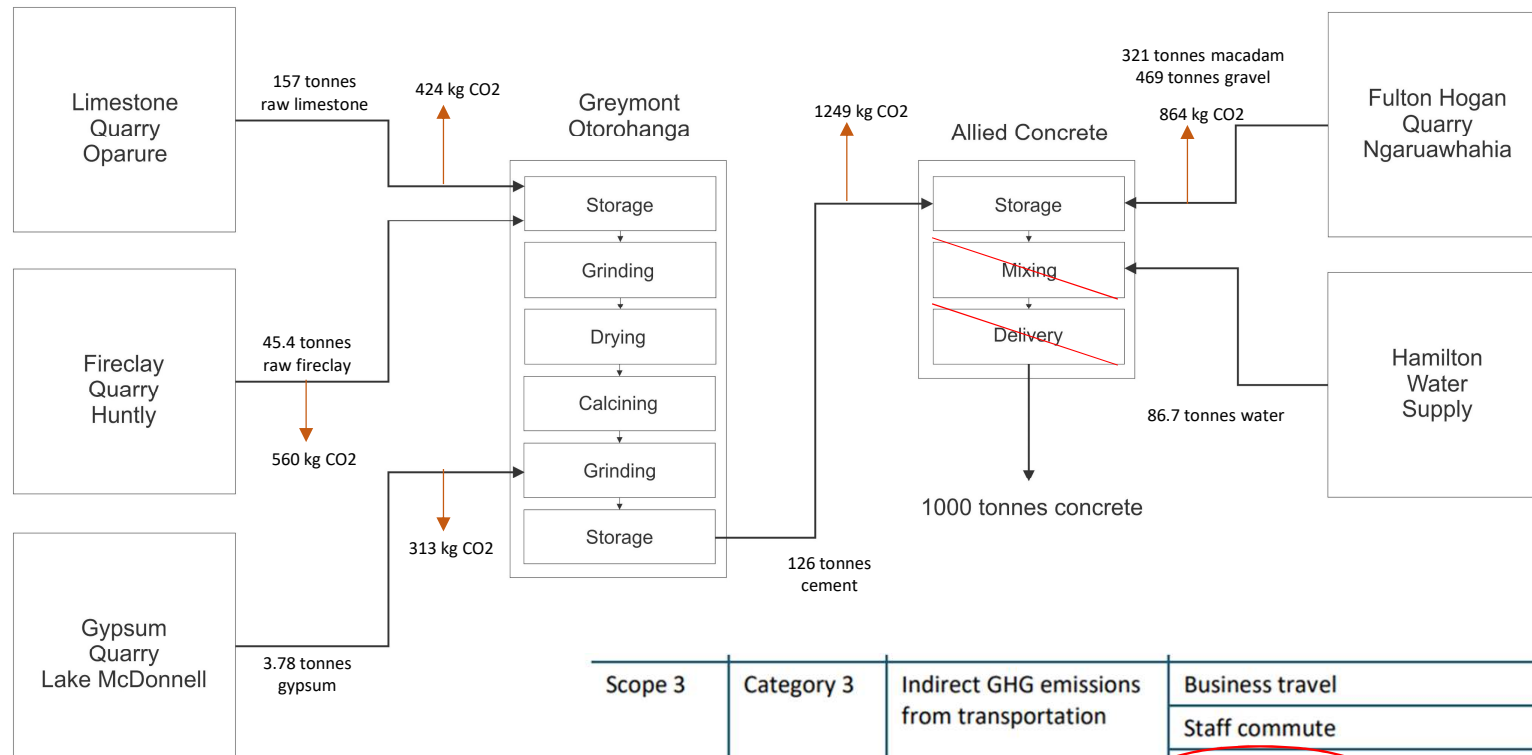
Scope 2 – Category 2



Scope 2	Category 2	Indirect GHG emissions from imported energy	Purchased energy
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Categorising emission sources

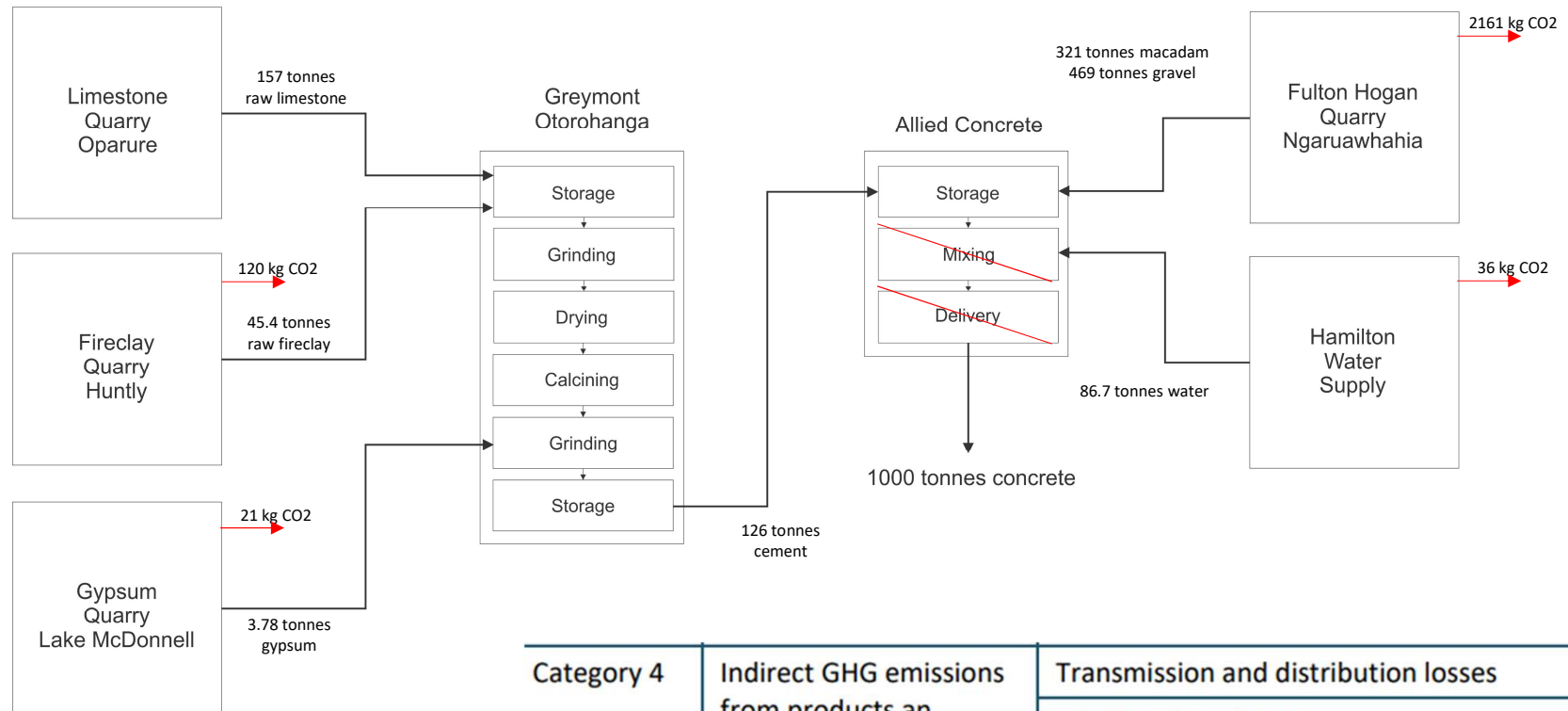
Scope 3 – Category 3



Scope 3	Category 3	Indirect GHG emissions from transportation	Business travel
			Staff commute
			Freight transport
			Refrigerant use (from chilled transport or air conditioner)

Categorising emission sources

Scope 3 – Category 4



Category 4	Indirect GHG emissions from products an organisation uses	Transmission and distribution losses
		Working from home
		Water supply and wastewater treatment
		Materials and waste

		CO2 emissions (kg)	% of grand total
Scope 1			
Limestone Quarry Oparure		430	0.22
Greymont Otorohanga	Drying	22378	11.6
	Calcining	163132	84.6
Total (Scope 1)		185940	96.5
Scope 2			
Greymont Otorohanga	Grinding 1	565	0.29
	Grinding 2	492	0.26
Total (Scope 2)		1057	0.55
Scope 3 - Category 3			
Limestone transport		424	0.22
Fireclay transport		560	0.29
Gypsum transport		313	0.16
Cement transport		1249	0.65
Macadam and gravel transport		864	0.45
Total (Scope 3 - Cat 3)		3410	1.77
Scope 3 - Category 4			
Fireclay		120	0.06
Gypsum		21	0.01
Macadam and gravel		2161	1.12
Water		36	0.02
Total (Scope 3 - Cat 3)		2338	1.21
Grand total		192745	

Conclusions

96.5% of the emissions are Scope 1 – emissions which the company has direct control over

0.55% of the emissions are Scope 2 – from the purchase of electricity

1.77% of the emissions are Scope 3, Category 3 – transport of materials

1.21% of the emissions are Scope 3, Category 4 – supply of materials

Limitations

- Neglected emissions
 - from materials transport on site – e.g. conveyors, vehicles
 - from transporting coal to Greymont
 - from mixing processes – e.g. premixing of powders before drying, mixing of gypsum with clinker before grinding, mixing of the concrete itself
 - from electricity required to run the kiln, drier and utilities
 - therefore emissions from electricity and fuel consumption will be slightly higher

Limitations

- Assumed quarrying operations are the same, regardless of location and product
 - The original quarry LCA had vehicles covering distances of 5 km
 - Oparure quarry size is about 0.5 km
 - Huntly quarry size is about 100-200 m
 - Fulton Hogan quarry is about 100-200 m
 - Gypsum quarry is about 6 km
 - Therefore vehicle use for Oparure, Huntly, and Fulton Hogan likely to be overestimated and emissions for material from those quarries likely to be less

But do the limitations really matter?

		CO2 emissions (kg)	% of grand total
Scope 1			
Limestone Quarry Oparure		430	0.22
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Grand total		192745	

No - not really

96% of the emissions are from the drying and calcining of the raw materials to make the cement

0.55% is from electricity consumption for grinding – the most electrically intensive part of the process, so you could expect electricity consumption for the other parts of the process to be trivial

1.21% + 0.22% of the emissions is from raw materials from all the quarries – so again any reduction in CO₂ emissions from these parts of the process would have a small impact on the overall emissions

Checking for consistency

- How do our results compare to other LCAs in similar areas?
- E.g. Electricity consumption – we calculated ~75 kWhr/tonne cement for grinding – but according to Stoiber 2003...

Table 2. Electrical energy consumption processes within a cement plant (Stoiber, 2003)

CONSUMER	ENERGY DEMAND (kWh/t)	%
Extraction & Blending	5.5	5
Raw material grinding	26.4	24
Raw meal homogenization	6.6	6
Clinker production	24.2	22
Cement grinding	41.8	38
Conveying, packing & etc	5.5	5
TOTAL	110	100

Overall 110 kWhr/tonne cement, so we need to multiply emissions from electricity consumption by ~1.5

We can also go back and refine the estimates and flow diagram for cement manufacture

Overall – it changes the CO₂ emissions per tonne of concrete from 192.7 to 193.3 – an overall change of 0.3%

Checking for consistency

- We neglected electricity consumption for conveying and mixing materials at Allied Concrete – using Stoiber 2003
 - Conveying and packing – 5.5 kWhr/tonne – 5500 kWhr
 - Blending – 5.5 kWhr/tonne – 5500 kWhr
 - This adds an extra 223 kg CO₂ , an additional 0.05%

Concrete mixtures

Materials	kg/tonne concrete		
	Our LCA	Colangelo et al 2018	Sjunnesson 2005
Cement	126	146	182
Macadam	321		399
Sand		391	
Gravel	469	391	347
Plasticiser (Peramin F)	0.65		0.40
Air entrainer (Paramin HPA)			1.38
Water	86.7	73	70
Notes	32 Mpa		52 MPa
	Impact (kg CO2/tonne concrete)		
	192	193	194

Emissions from concrete production largely insensitive to concrete composition

Overall comparison

LCA	Type	kg CO2/tonne
Our LCA	Concrete (32 MPa)	192
Colangelo et al 2018		310
Nebel et al 2011	Concrete (17.5 Mpa)	102
Nebel et al 2011	Concrete (40 Mpa)	158
Nebel et al 2011	Pre-Cast Concrete	170
Nebel et al 2011	Concrete tiles	263
Mohammadi and South 2017		87-217
Sjunnesson 2005	Concrete (32 Mpa)	107.5
Sjunnesson 2005	Concrete (52 Mpa)	150.8

Take home messages

- Categorising emissions according to Scopes helps identify where the bulk of the emissions come from and what emissions you have direct control over
 - Be very explicit in stating your assumptions
- Good to review limitations of your LCA as part of the discussion
 - But don't be negative, assess the limitations for their likely impact on the overall LCA results, draw useful conclusions
 - In the case of the concrete LCA the limitations probably had a negligible effect

Take home messages

- Good to compare your results with other LCAs
 - The concrete LCA underestimated electricity consumption at the cement factory
 - The LCA was largely insensitive to concrete formulation (within the range tried)
 - The LCA was within the range of results from other LCAs on concrete
- You can be infinitely thorough and take infinite time, but it will only affect the final impact result by 1-2%

LCA 4.2 – Sensitivity Analysis

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Sensitivity analysis

- Involves taking key parameters: compositions, mass flows, energy needs and impacts and adjusting each one sequentially up and down e.g. by 10%, and examine the effect of that parameter on the overall environmental impact
- Some LCAs use greater ranges (e.g. 20-50%) for some parameters to reflect the greater variation or uncertainty for those particular parameters

Why conduct a sensitivity analysis?

- To determine which aspects of a process, system, product, infrastructure you should focus on to reduce the overall environmental impact
- To test your LCA model, make sure it is behaving, and you do not get weird results
 - E.g. the environmental impact does not update when you change a parameter, or the change is smaller or bigger than you might expect
 - Might be broken (or no) links between Excel cells
 - Some of the calculations may have been done manually and you need to put in a formula in your Excel spreadsheet

Why conduct a sensitivity analysis?

- For simple LCAs, e.g. in Excel, you already have a good idea of what the key parameters are
- For complex LCAs using LCA software, e.g. Gabi, considering a wide range of impacts, using impact databases (e.g. Ecoinvent) which has 100's of impacts per material

When to do it?

- Basically after you have completed your LCA model and troubleshooted it as best you can

Basic set up

Base case (kg CO2 emissions/tonne concrete)	192.8	Result of variation (kg CO2 emissions/tonne concrete)	192.8
		Variation (%)	10

Materials	Concrete composition	
	kg in mixture	mass fraction
Cement	126	0.126
Macadam	321	0.320
Gravel	469	0.467
Plasticiser	0.65	0.001
Water	86.7	0.086
Total (kg)	1003.35	

Materials	Concrete composition (kg in mixture)			kg CO2 emissions/tonne concrete)		% difference from base case	
	Base case	Low	High	Low	High	Low	High
Cement	126						
Macadam	321						
Gravel	469						
Plasticiser	0.65						
Water	86.7						

Sensitivity results – concrete and cement composition

	Concrete composition (kg in mixture)			kg CO2 emissions/tonne concrete)		% difference from base case	
Materials	Base case	Low	High	Low	High	Low	High
Cement	126	113.4	138.6	176.0	209.1	-8.70	8.48
Macadam	321	288.9	353.1	199.0	186.9	3.24	-3.04
Gravel	469	422.1	515.9	202.1	184.4	4.80	-4.38
Plasticiser	0.65	0.585	0.715				
Water	86.7	78.03	95.37	194.5	191.2	0.87	-0.86
Total (kg)							
	Cement composition (kg in mixture)			kg CO2 emissions/tonne		% difference from base case	
Material	Base case	Low	High	Low	High	Low	High
Calcium oxide, CaO	670	603	737	189.0	196.2	-1.99	1.74
Fireclay	300	270	330	195.9	189.8	1.63	-1.54
Gypsum	30	27	33	193.3	192.3	0.28	-0.28

Sensitivity results – concrete and cement composition

- But hang on – you said...

Materials	kg/tonne concrete		
	Our LCA	Colangelo et al 2018	Sjunnesson 2005
Cement	126	146	182
Macadam	321		399
Sand		391	
Gravel	469	391	347
Plasticiser (Peramin F)	0.65		0.40
Air entrainer (Paramin HPA)			1.38
Water	86.7	73	70
Notes	32 Mpa		52 MPa
	Impact (kg CO2/tonne concrete)		
	192	193	194

The concrete LCA had a couple of missing links for mass flows, so one part updated, the other part did not

Emissions from concrete production largely insensitive to concrete composition

Sensitivity results – concrete and cement composition

- Corrected numbers

Materials	kg/tonne concrete		
	Our LCA	Colangelo et al 2018	Sjunnesson 2005
Cement	126	146	182
Macadam	321		399
Sand		391	
Gravel	469	391	347
Plasticiser (Peramin F)	0.65		0.40
Air entrainer (Paramin HPA)			1.38
Water	86.7	73	70
Notes	32 Mpa		52 MPa
	Impact (kg CO2/tonne concrete)		
	192	193	194
		223.3	278.3

LCA	Type	kg CO2/tonne
Our LCA	Concrete (32 MPa)	192
Colangelo et al 2018		310
Nebel et al 2011	Concrete (17.5 Mpa)	102
Nebel et al 2011	Concrete (40 Mpa)	158
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Sjunnesson 2005	Concrete (32 Mpa)	107.5
Sjunnesson 2005	Concrete (52 Mpa)	150.8

Sensitivity results – limestone and fireclay composition

	Limestone reaction and properties			kg CO2 emissions/tonne		% difference from base case	
Limestone reaction and properties	Base case	Low	High	Low	High	Low	High
Calcium oxide produced per tonne limestone	0.56	0.50	0.62	212.2	177.1	10.07	-8.12
Calcium carbonate content of limestone	0.96	0.86	1.06	211.6	177.5	9.75	-7.92
	Fireclay properties			kg CO2 emissions/tonne		% difference from base case	
Fireclay properties	Base case	Low	High	Low	High	Low	High
Fireclay content of rock	0.832	0.75	0.92	194.7	191.2	0.96	-0.84

Sensitivity results – Transport

	Transport gCO ₂ /tonne.km			kg CO ₂ emissions/tonne		% difference from base case	
Transport	Base case	Low	High	Low	High	Low	High
Rail	28	25.20	30.80	192.8	192.8	0.00	0.00
Small container vessel	12	10.80	13.20	192.8	192.8	-0.01	0.01
Road transport	135	121.50	148.50	192.5	193.1	-0.17	0.16
	Transport distance (km)			kg CO ₂ emissions/tonne		% difference from base case	
Transport distances	Base case	Low	High	Low	High	Low	High
Oparure Quarry - Graymont, Otorohanga	20	18	22	192.8	192.8	-0.02	0.02
Huntly Quarry - Graymont, Otorohanga	91.3	82	100	192.7	192.9	-0.03	0.03
Lake McDonnell - Thevenard	64	58	70	192.8	192.8	0.00	0.00
Thevenard - Port of Tauranga	5126	4614	5639	192.8	192.8	-0.01	0.01
Port of Tauranga - Graymont, Otorohanga	143	129	157	192.8	192.8	0.00	0.00
Graymont, Otorohanga - Allied Concrete	73.4	66	81	192.7	192.9	-0.07	0.07
Fulton Hogan, Ngaruawahia - Allied Concrete	8.1	7	9	192.7	192.9	-0.06	0.05

Sensitivity results – Environmental impacts from materials and electricity use

Material	Material impact (kg CO2/tonne)			kg CO2 emissions/tonne		% difference from base case	
	Base case	Low	High	Low	High	Low	High
Limestone	2.74	2.46	3.01	192.8	192.8	-0.02	0.02
Fireclay	2.74	2.46	3.01	192.8	192.8	-0.01	0.01
Gypsum	5.50	4.95	6.05	192.8	192.8	0.00	0.00
Aggregate and gravel	2.74	2.46	3.01	192.6	193.0	-0.11	0.11
Water	0.42	0.38	0.46	192.8	192.8	0.00	0.00
	Emissions from electricity, NZ			kg CO2		% difference from	
	Base case	Low	High	Low	High	Low	High
Emissions from electricity, NZ (kg CO2/MJ)	0.031	0.028	0.034	192.7	192.9	-0.05	0.05
Electricity consumption	Power (kWhr/tonne)			kg CO2		% difference from	
	Base case	Low	High	Low	High	Low	High
Grinding limestone and clay	25	22.5	27.5	192.7	192.9	-0.03	0.03
Grinding the clinker and gypsum	35	31.5	38.5	192.8	192.9	-0.02	0.03

Sensitivity results – Water content and heat capacity

	Water content			kg CO2 emissions/tonne		% difference from base case	
Material	Base case	Low	High	Low	High	Low	High
Limestone	0.02	0.018	0.022	193.0	192.6	0.11	-0.11
Fireclay	0.168	0.25	0.08	194.7	191.2	1.0	-0.8
	Heat capacity (kJ/kg.K)			kg CO2 emissions/tonne		% difference from base case	
Material	Base case	Low	High	Low	High	Low	High
Limestone	0.909	0.818	1.000	191.3	194.3	-0.78	0.78
Fireclay	0.878	0.790	0.966	192.4	193.2	-0.18	0.19

Sensitivity results – Dryer operating conditions, coal, and calciner heating needs

Parameter	Value			kg CO2 emissions/tonne		% difference from base case	
	Base case	Low	High	Low	High	Low	High
Starting dryer temperature (oC)	25	22.5	27.5	192.9	192.7	0.04	-0.04
Final drying temperature (oC)	800	720	880	190.9	194.7	-0.99	0.99
Efficiency of drier	0.8	0.72	0.88	195.3	190.8	1.29	-1.05
Coal CV (MJ/kg)	30	27	33	205.9	182.1	6.79	-5.56
CO2 emissions burning coal (kg CO ₂ /kg coal)	3.32	2.99	3.65	181.1	204.5	-6.07	6.08
Calciner heat required (MJ heat/kg feed)	4.5	4.05	4.95	183.3	202.3	-4.95	4.95

Conclusions from the sensitivity analysis

- The environmental impact of concrete is very sensitive ($\pm 5-9 \%$) to
 - Concrete composition (i.e. the amount of cement in it)
 - Calcium oxide produced per tonne of limestone
 - Amount of calcium carbonate per tonne of limestone
 - Calorific value of the coal
 - Emissions from burning the coal
 - Calciner heat required
- The environmental impact of concrete is sensitive ($\pm 1-2 \%$) Cement composition
 - Dryer efficiency
 - Water content of fireclay

Conclusions from the sensitivity analysis

- The environmental impact of concrete is not sensitive to:
 - Gypsum content in the cement
 - Water content of limestone (low anyway)
 - Heat capacity of fireclay
 - Starting drier temperature
 - Transportation emissions and transportation distances
 - Electricity consumption

Conclusions from the sensitivity analysis

- To reduce the environmental impact of concrete you need to:
 - Reduce the amount of cement needed for the same strength concrete
 - Reduce the amount of calcium oxide required for cement
 - Use limestone with a high calcium carbonate content and low water content
 - Ensure good conversion of calcium carbonate to calcium oxide
 - Reduce the water content of fireclay rock
 - Use a limestone with a lower heating capacity (if possible)
 - Use a lower drying temperature and high efficiency drier
 - Use a fuel with high calorific value and low CO₂ emissions
 - Use an efficient calciner with a low heating requirement

Take home messages

- Sensitivity analysis is great for trouble shooting
- It is great for being able to analyse in a quantitative manner the effect of each parameter on your LCA
- Gives you something to talk about in the discussion for your LCA